

Fig 5 — Accumulator Precision Recommendation Table
Minimum sufficient accMantBits P* per config and K (normErr ≤ 110% of FP32 baseline; K ∈ {4, 8, 16, 32, 64})

Config	Class	rq→	K=4	K=8	K=16	K=32	K=64	Rec. accReg	Saves vs FP32	normFloor%	K sensitivity
INT8xINT8 N4	ClassA	INT8	7	7	7	9	9	FP18 (9-bit M)	−14 bits	3.47%	K≤4→7b
INT8xINT8 N8	ClassA	INT8	7	7	7	7	9	FP18 (9-bit M)	−14 bits	3.41%	K≤4→7b
INT8xE4M3 N4	ClassB	E4M3	7	7	7	7	7	FP16 (7-bit M)	−16 bits	6.05%	flat
INT8xE4M3 N8	ClassB	E4M3	7	7	7	7	9	FP18 (9-bit M)	−14 bits	6.17%	K≤4→7b
E4M3xE4M3 N4	ClassC	E4M3	7	7	7	7	7	FP16 (7-bit M)	−16 bits	6.42%	flat
E4M3xE4M3 N8	ClassC	E4M3	7	7	7	7	9	FP18 (9-bit M)	−14 bits	6.50%	K≤4→7b
E5M2xE5M2 N4	ClassC	E5M2	7	7	9	7	12	FP21 (12-bit M)	−11 bits	8.82%	K≤4→7b
E5M2xE4M3 N4	ClassC	E4M3	7	7	7	7	7	FP16 (7-bit M)	−16 bits	6.13%	flat
INT8xE2M1 N4	INT8xFP4	INT8	7	7	7	9	9	FP18 (9-bit M)	−14 bits	3.51%	K≤4→7b
INT8xE2M1 N8	INT8xFP4	INT8	7	7	7	7	9	FP18 (9-bit M)	−14 bits	3.27%	K≤4→7b
E4M3xE2M1 N4	FP8xFP4	E4M3	7	7	7	9	7	FP18 (9-bit M)	−14 bits	5.99%	K≤4→7b
E4M3xE2M1 N8	FP8xFP4	E4M3	7	7	7	7	9	FP18 (9-bit M)	−14 bits	6.50%	K≤4→7b
E5M2xE2M1 N4	FP8xFP4	E4M3	7	7	7	7	9	FP18 (9-bit M)	−14 bits	6.52%	K≤4→7b

Colour key: blue=ClassA (INT8×INT8), orange=ClassB (INT8×FP8), green=ClassC (FP8×FP8), red=INT8×FP4, purple=FP8×FP4. Rec. accReg = worst-K P* → design-time Chisel parameter. normFloor% = mean normalised error with FP32 accumulator (pure requant noise).